Harsh Shah

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Education

Master of Science in Data Science – University of the Pacific, **San Francisco, CA, USA** Aug 2021 - May 2023
Bachelor of Technology in Computer Science – Gujarat Technological University, India Aug 2015 - May 2019

Technical Skills

Programming Languages: Python, SQL, C/C++, R, Java
Frameworks & Libraries: Pandas, NumPy, Langchain, Scikit-learn, TensorFlow, Beautiful Soup, Matplotlib, Seaborn, Plotly
Databases & Tools: MS SQL Server, MongoDB, Pinecone, Git, Tableau
Cloud & MLOps: AWS (EC2, S3, SageMaker)

Experience

Research Intern - Generative AI, AI Institute of South Carolina - Remote, India Nov 2024 - March 2025

- Annotated and classified **over 2,000 text-to-image samples** using Vision Language Models, improving **anomaly detection** workflows and reducing annotation time significantly.
- Developed and refined **annotation protocols** leveraging agentic frameworks for AI-driven labeling, boosting accuracy by 30% and strengthening reliability for **hallucination detection** in generative models.

Data Science Intern, Flexon Technologies Inc. **– Pleasanton, CA, USA** Feb 2024 – Nov 2024

- Developed a scalable LLM-powered chatbot for the generation of job descriptions using **chain of thought prompting with GPT-4**, significantly reducing manual input effort.
- Designed and implemented a modular, agent-based architecture with **Retrieval-Augmented Generation (RAG)**, integrating **Pinecone** and **LangChain** for efficient semantic retrieval and improving contextual accuracy.
- Built and automated an end-to-end **data pipeline** to parse resumes and classify candidate profiles.
- Defined and implemented **evaluation metrics for LLM output**, including precision, recall, and bias checks, and added content filtering layers to ensure safe and compliant deployment of generated content.

Data Science Intern, QQ Tech, Inc.(Startup) **– San Francisco, USA** Jan 2023 – May 2023

- Designed a web crawler with **Beautiful Soup (Python)**, extracting 50,000+ HTML files at 33 pages per minute.
- Designed and trained a **text classification model** using a deep learning architecture, achieving an accuracy of 92% in classifying HTML files into 'good' and 'non-relevant' categories, reducing manual review efforts significantly.
- Reduced data errors through advanced cleaning techniques like **duplicate-row elimination, schema validation with pandas, and automated null-value imputation**, improving overall data quality.
- Contributed to improvement in project delivery timelines through effective collaboration in stand-ups, code reviews, and brainstorming sessions, fostering a more efficient and cohesive team environment.

Projects

Personalized Product Recommendation Platform (*Flask, Python, MongoDB, OpenAI, Tailwind/Bootstrap*) github

- Designed and built a **full-stack AI-powered product recommendation system** that delivers daily personalized app/tool suggestions to users via email, leveraging OpenAI embeddings and advanced hybrid scoring algorithms.
- Constructed an automated data enrichment and feature engineering pipeline, integrating with **LLM APIs (OpenAI, Perplexity)** to process raw product data, generate high-dimensional vector embeddings, and store them in **MongoDB** for efficient similarity search.

Customer Segmentation using Credit Card Data (*Python, Pandas, Scikit-learn, Matplotlib*) github

- Conducted customer segmentation analysis using credit card transaction data, achieving a 10–15% gain in targeted marketing effectiveness while leveraging scalable data processing techniques.
- Utilized k-means and hierarchical clustering to group customers by spending behavior, enabling personalized engagement and integration with ML model evaluation metrics.

Telecom Customer Churn Prediction (*Python, Pandas, Scikit-learn, Matplotlib*) github

- Engineered a churn prediction system with 82% accuracy using advanced feature engineering and model tuning, integrating scalable APIs and microservices.
- Leveraged data visualization for stakeholder insights, identifying key churn factors and applying ML model fine-tuning to ensure consistent performance.

Global Health Expenditure Profile (*Tableau, Data Visualization, Storytelling*)

- Created an interactive Tableau dashboard showcasing global health expenditure trends, improving data analysis speed, and enhancing understanding of global health economics.